

Lecture 4: Divisibility and the Euclidean Algorithm

Course 21-127 — Concepts of Mathematics

Week 2

Contents

1	Divisibility	1
2	The division theorem	1
3	Greatest common divisors	2

1 Divisibility

Definition 1.1 (Divides). Let $a, b \in \mathbb{Z}$. We say that a divides b , written $a \mid b$, if there exists $k \in \mathbb{Z}$ such that $b = ka$. If no such k exists we write $a \nmid b$.

Example 1.2. $3 \mid 12$ since $12 = 4 \cdot 3$, but $5 \nmid 12$. Every integer divides 0, and 1 divides every integer.

Lemma 1.3. If $a \mid b$ and $b \mid c$ then $a \mid c$.

Proof. Write $b = ka$ and $c = \ell b$ for some $k, \ell \in \mathbb{Z}$. Then $c = \ell b = \ell(ka) = (\ell k)a$, and $\ell k \in \mathbb{Z}$, so $a \mid c$. $\square$

Lemma 1.4. If $a \mid b$ and $a \mid c$ then $a \mid (mb + nc)$ for all $m, n \in \mathbb{Z}$.

Proof. Write $b = ka$ and $c = \ell a$. Then $mb + nc = mka + n\ell a = (mk + n\ell)a$. $\square$

Remark. Lemma 1.4 is the fact that makes the Euclidean algorithm work: the set of common divisors of a and b is unchanged when b is replaced by $b - qa$.

2 The division theorem

Theorem 2.1 (Division with remainder). Let $a \in \mathbb{Z}$ and $b \in \mathbb{Z}_{>0}$. There exist unique $q, r \in \mathbb{Z}$ with

$$a = qb + r \quad \text{and} \quad 0 \leq r < b.$$

Proof. Existence. Consider the set $S = \{a - kb : k \in \mathbb{Z}, a - kb \geq 0\}$. The set is non-empty (take $k = -|a|$), so by well-ordering it has a least element $r = a - qb$. If $r \geq b$ then $r - b = a - (q+1)b \in S$ is smaller than r , a contradiction. Hence $0 \leq r < b$.

Uniqueness. Suppose $a = qb + r = q'b + r'$ with $0 \leq r, r' < b$. Then $(q - q')b = r' - r$ and $|r' - r| < b$, so $b \mid (r' - r)$ forces $r' - r = 0$, and therefore $q = q'$. $\square$

Corollary 2.2. Every integer is either even or odd, and not both.

3 Greatest common divisors

Definition 3.1. For $a, b \in \mathbb{Z}$ not both zero, $\gcd(a, b)$ is the largest $d \in \mathbb{Z}_{>0}$ with $d \mid a$ and $d \mid b$. We also write $\text{lcm}(a, b)$ for the least positive common multiple.

Theorem 3.2 (Euclidean algorithm). *Let $a \geq b > 0$ and write $a = qb + r$ as in Theorem 2.1. Then $\gcd(a, b) = \gcd(b, r)$.*

Proof. By Lemma 1.4, every common divisor of a and b divides $r = a - qb$, and every common divisor of b and r divides $a = qb + r$. The two pairs therefore have the same common divisors, hence the same greatest one. $\square$

Example 3.3. To compute $\gcd(252, 198)$:

$$252 = 1 \cdot 198 + 54,$$

$$198 = 3 \cdot 54 + 36,$$

$$54 = 1 \cdot 36 + 18,$$

$$36 = 2 \cdot 18 + 0.$$

The last non-zero remainder is 18, so $\gcd(252, 198) = 18$.

Theorem 3.4 (Bézout). *For $a, b \in \mathbb{Z}$ not both zero there exist $m, n \in \mathbb{Z}$ with $ma + nb = \gcd(a, b)$.*

Proof sketch. Run the Euclidean algorithm and substitute backwards; each remainder is an integer combination of the previous two, hence of a and b . $\square$

Exercises

- (i) Prove that $\gcd(n, n + 1) = 1$ for every $n \in \mathbb{N}$.
- (ii) Find $m, n \in \mathbb{Z}$ with $252m + 198n = 18$.
- (iii) Show that $\gcd(a, b) \cdot \text{lcm}(a, b) = ab$ for $a, b \in \mathbb{Z}_{>0}$.